

Overview and Biological Roles:

Vitamin B12 (cobalamin) is a water-soluble vitamin that is naturally present in some foods, added to others, and available as a dietary supplement. Vitamin B12 is essential for:

- Red blood cell formation and erythropoiesis
- Neurological function and maintenance of the myelin sheath
- DNA synthesis and homocysteine regulation (working as a cofactor with folate)

Recommended Dietary Allowances (RDA) for Vitamin B12:

The Food and Nutrition Board established the following daily RDAs:

- Infants 0-6 months: 0.4 mcg (Adequate Intake)
- Infants 7-12 months: 0.5 mcg (Adequate Intake)
- Children 1-3 years: 0.9 mcg/day
- Children 4-8 years: 1.2 mcg/day
- Children 9-13 years: 1.8 mcg/day
- Adolescents and Adults 14+ years: 2.4 mcg/day
- Pregnant Females: 2.6 mcg/day
- Lactating Females: 2.8 mcg/day

Tolerable Upper Intake Level (UL):

The FNB has not established a UL for vitamin B12 because of its low potential for toxicity. Excess amounts are typically excreted in urine.

Food Sources of Vitamin B12:

Vitamin B12 is naturally found primarily in foods of animal origin:

- Fish and Seafood: Clams, oysters, salmon, trout, and tuna.
- Meat and Poultry: Beef liver, beef, and turkey.
- Dairy and Eggs: Milk, yogurt, cheese, and eggs.
- Fortified Foods: Fortified breakfast cereals and nutritional yeasts are critical sources for vegetarians and vegans. Unfortified plant foods contain no vitamin B12.

Absorption Mechanisms:

Absorption of vitamin B12 requires gastric hydrochloric acid to release the vitamin from food protein, and intrinsic factor (secreted by gastric parietal cells) to facilitate ileal absorption. Individuals with atrophic gastritis, pernicious anemia, or previous gastric bypass surgery have impaired absorption.

Deficiency Manifestations:

Vitamin B12 deficiency leads to megaloblastic anemia, fatigue, weakness, constipation, and loss of appetite. Neurological symptoms include numbness and tingling in the hands and feet (peripheral neuropathy), difficulty walking, balance problems, memory loss, and depression. Neurological damage can become permanent if untreated.